

Edward Njiru - Data Scientist

WHERE IS THE MONEY?



A Time-Series Analysis on Agricultural
Commodity Market Volumes and Prices
in Kenya (Q3/2021 - Q2/2024)

Part II: Perennial Crops

INTRO



Hello! My name is Edward Njiru, and I have been a dairy and fodder farmer in Kenya for the last four years.

Like several farmers, the decision to diversify is not an easy one. There is so much to consider regarding demand and profit margins. So I took a deep-dive into the data.

Part II: *Perennial Crops*

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DATA



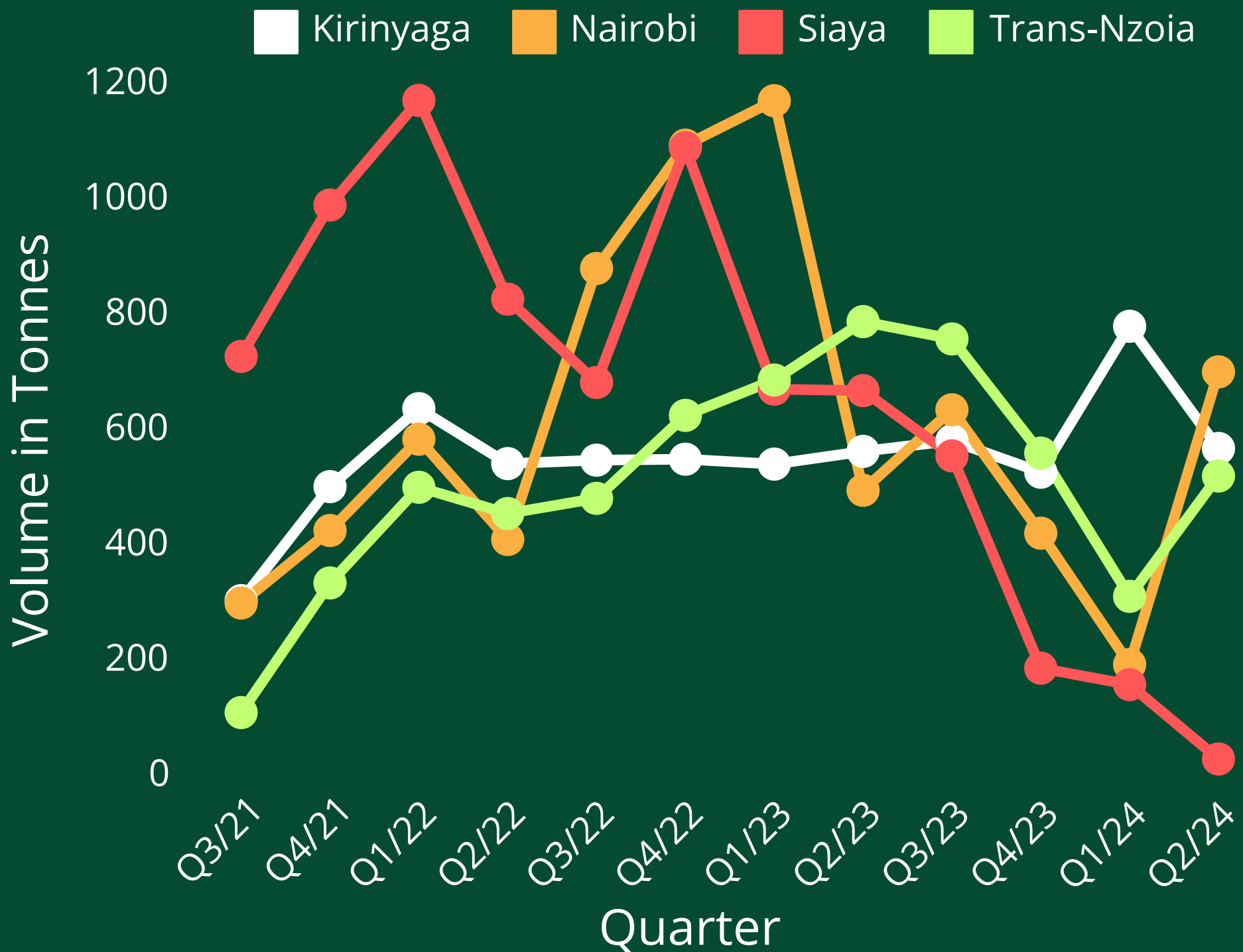
The data was sourced from the Kenya Agricultural Market Information System (KAMIS). The Objective Questions:

1. Where is the demand for commodities?
2. Which commodities are in demand?
3. Which commodities are most profitable?
4. Which profitable commodities are in demand?

TOP COUNTIES & MARKETS

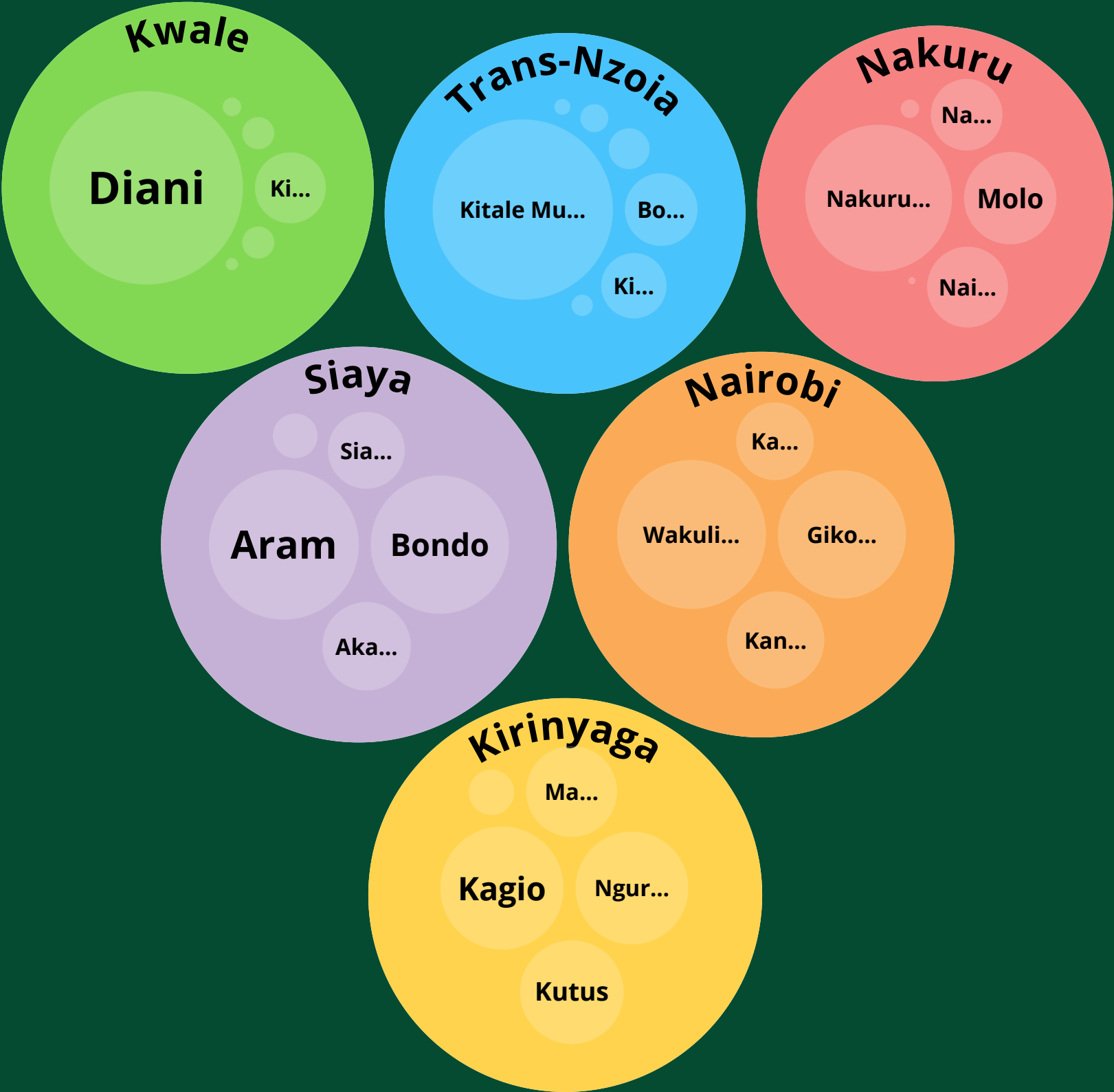
DEMAND BY VOLUME IN TONNES

Top 4 counties based on sales volume in markets.



TOP COUNTIES & MARKETS

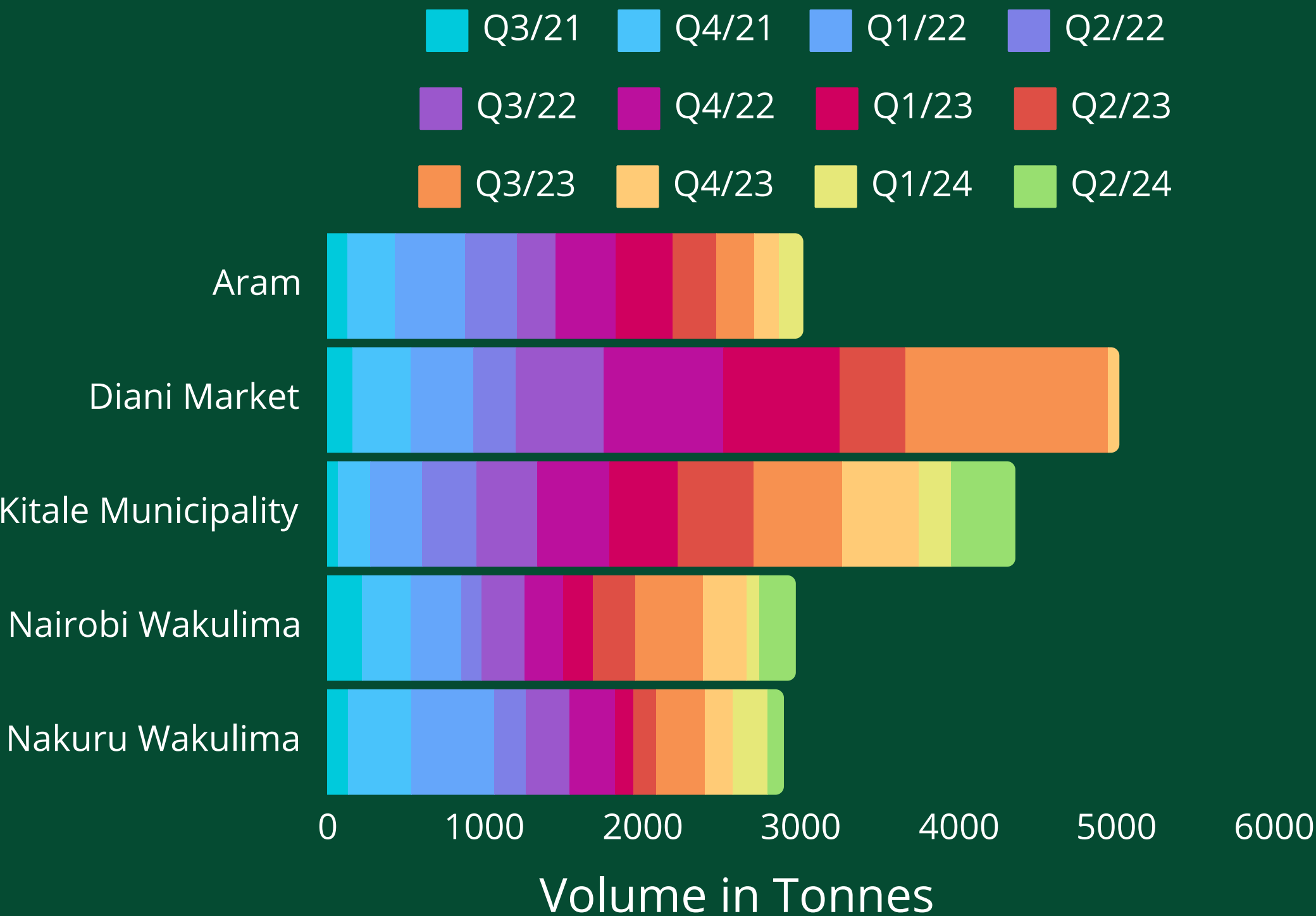
DEMAND BY VOLUME IN TONNES



TOP COUNTIES & MARKETS

DEMAND BY VOLUME IN TONNES

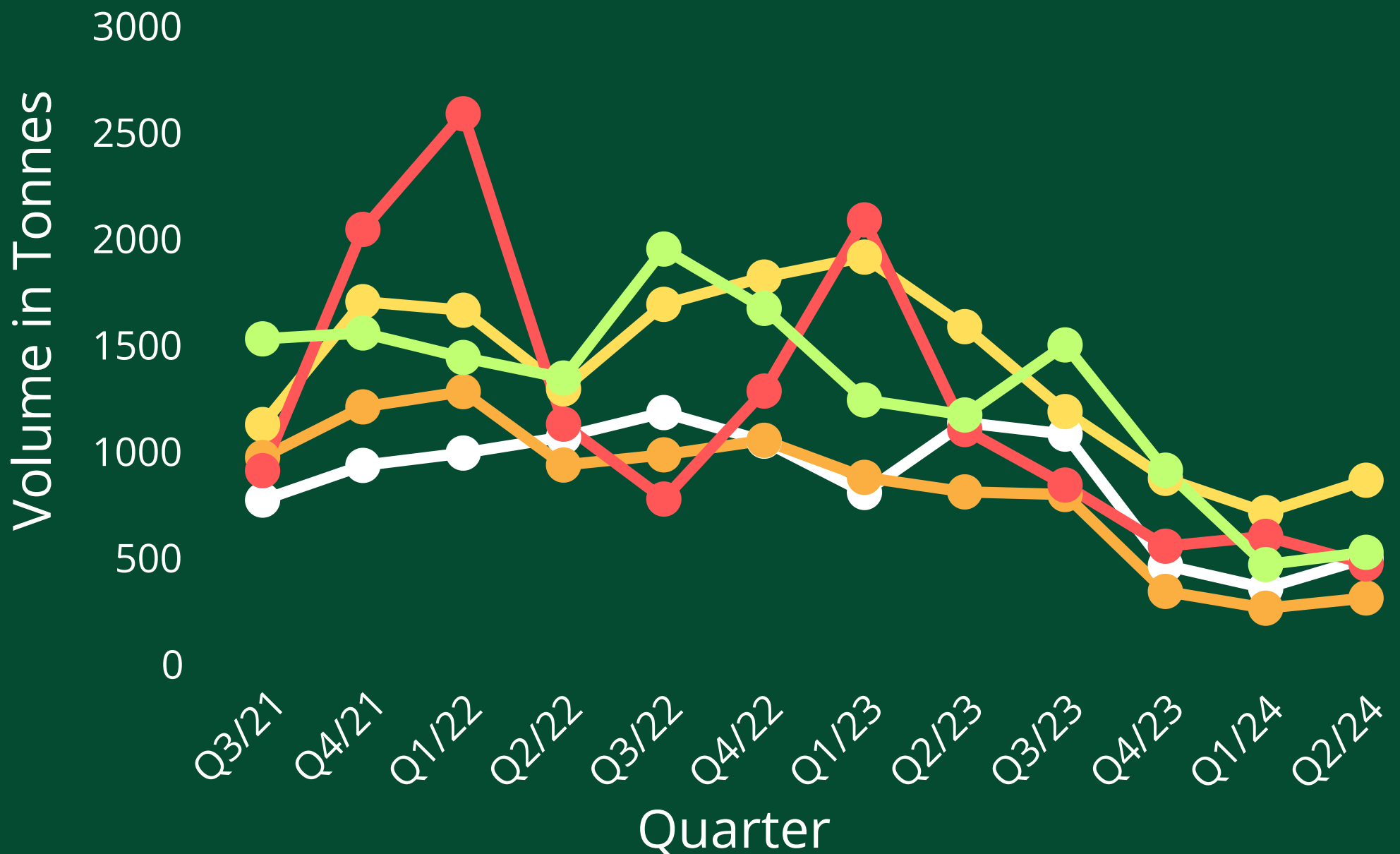
Top 5 food markets based on sales volume.



TOP COMMODITIES

DEMAND BY VOLUME IN TONNES

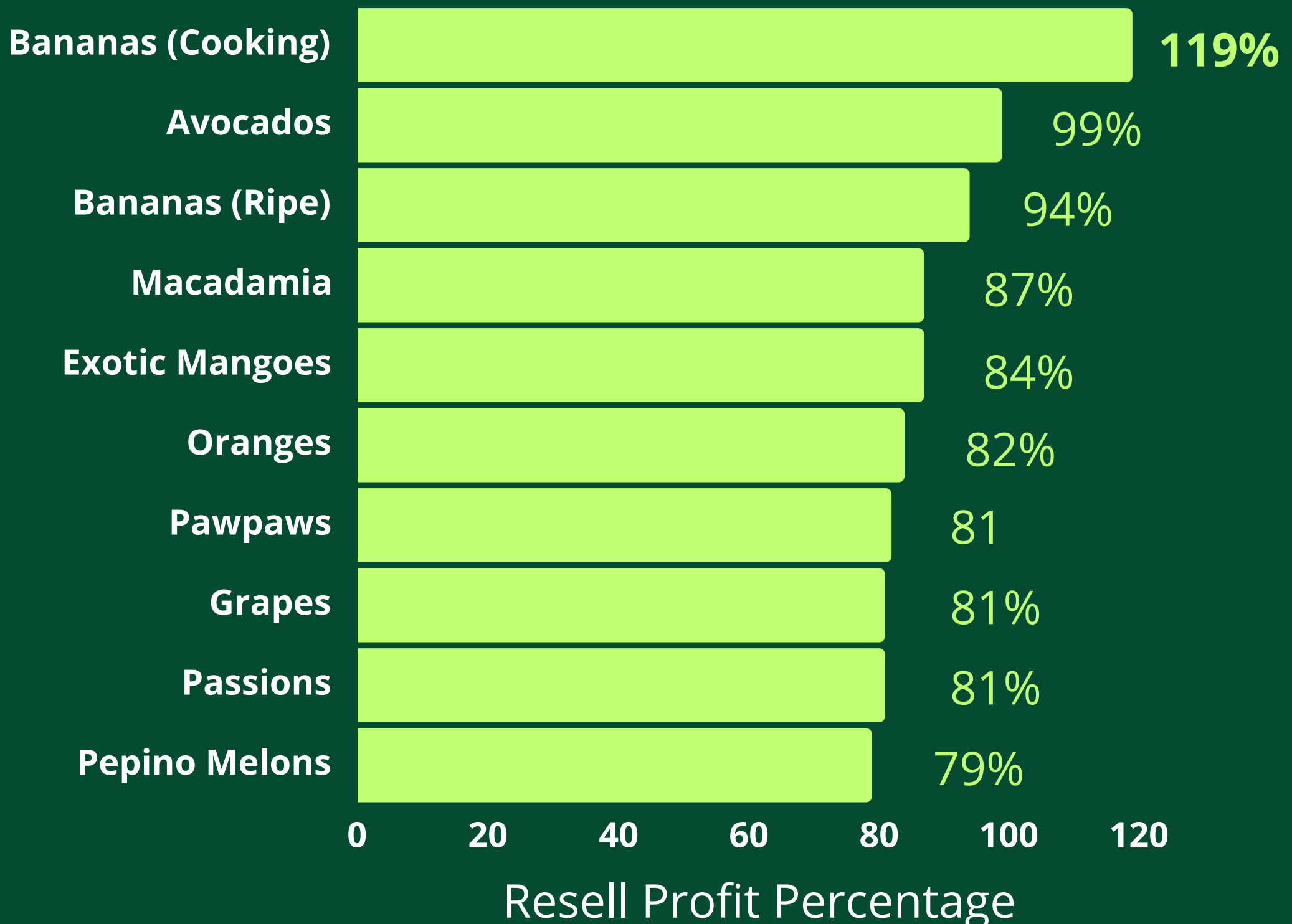
Top 5 commodities based on sales volume in markets.



TOP COMMODITIES

AVERAGE RESELL PROFIT %

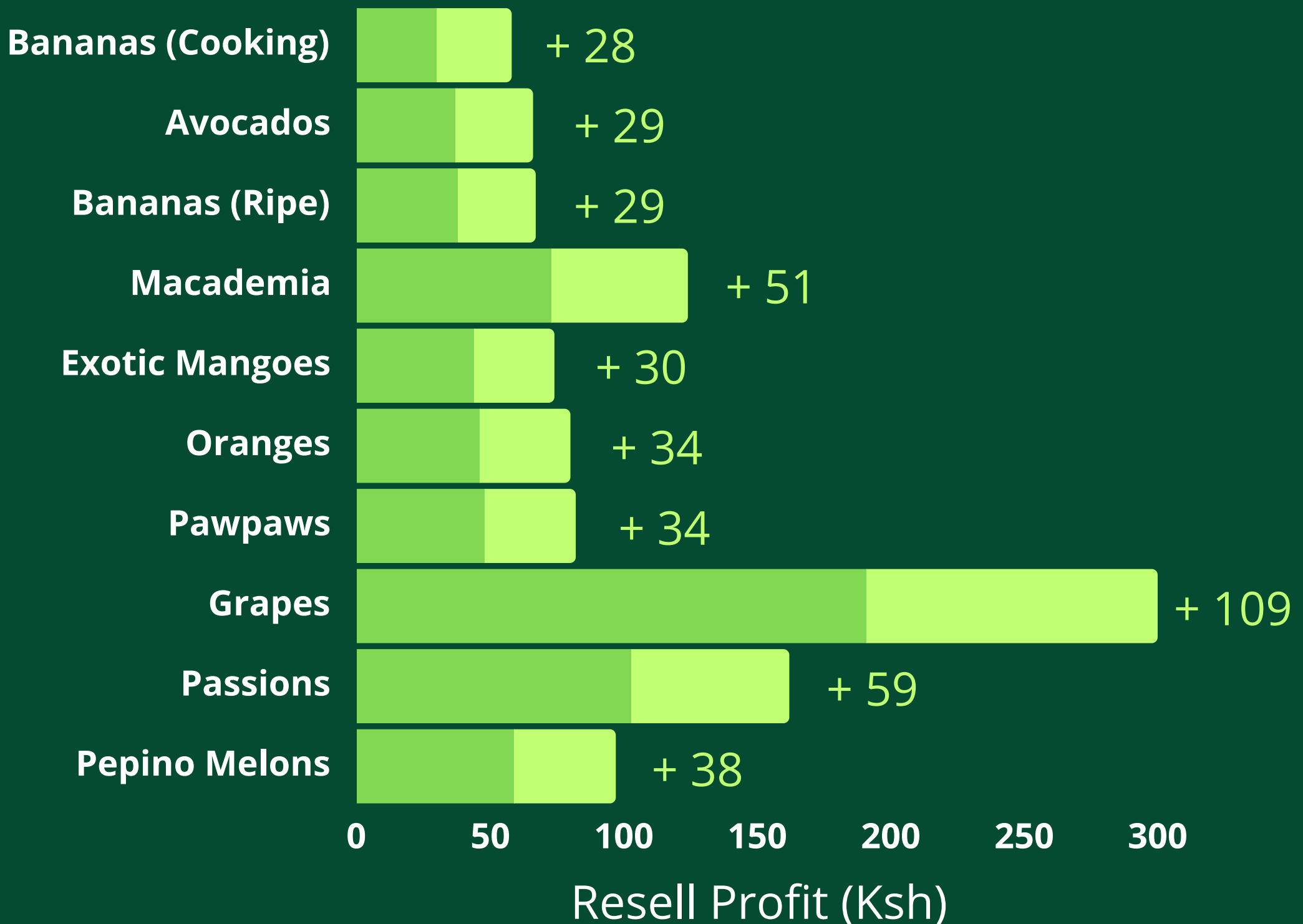
Most profitable commodities for resell.



TOP COMMODITIES

AVERAGE RESELL PROFIT MARGINS

Most profitable commodities for resell.



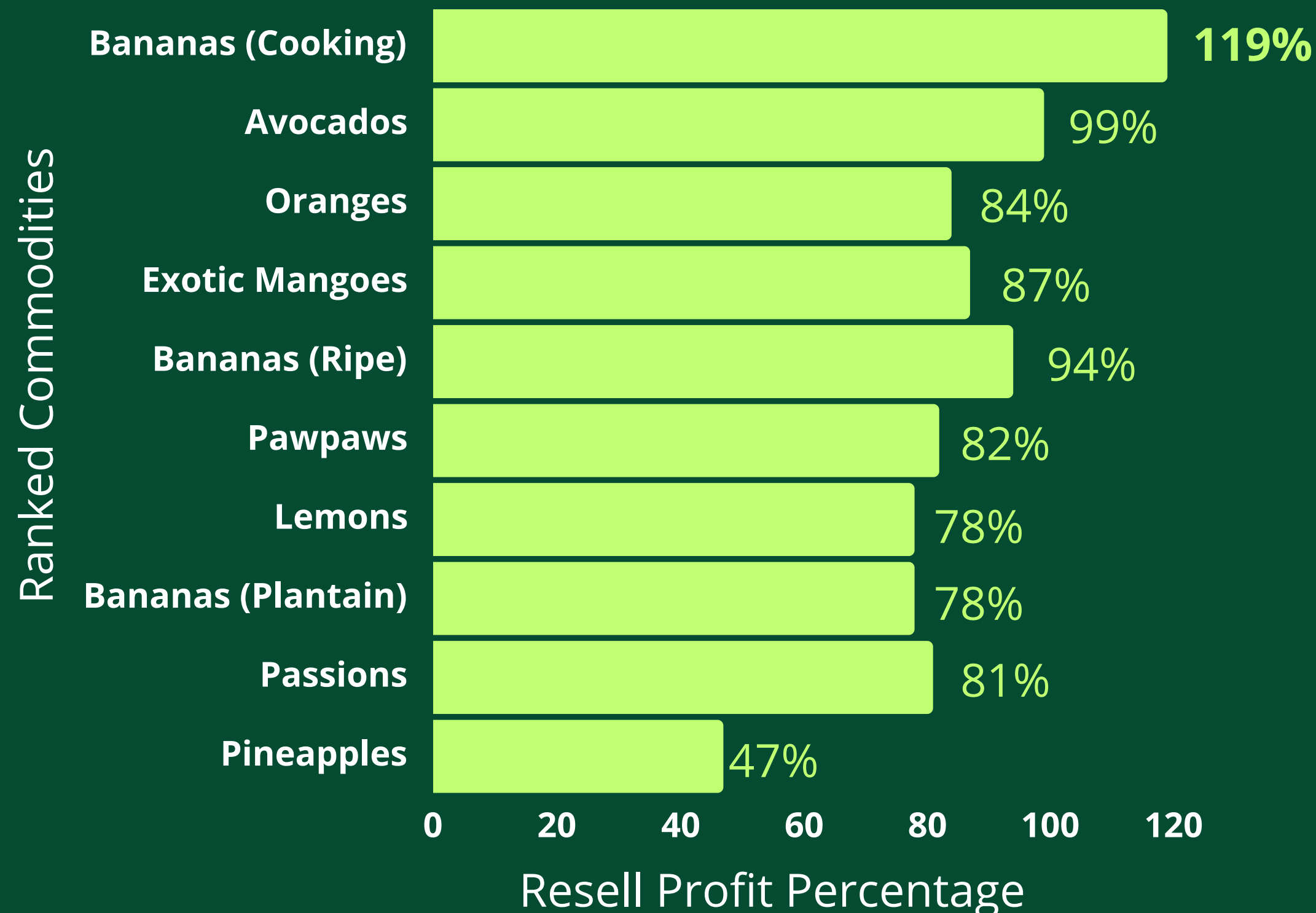
It is expected that niche crops have excellent margins.

As a farmer, I would want to know which commodities have **high demand** **AND** **good profit margins**.



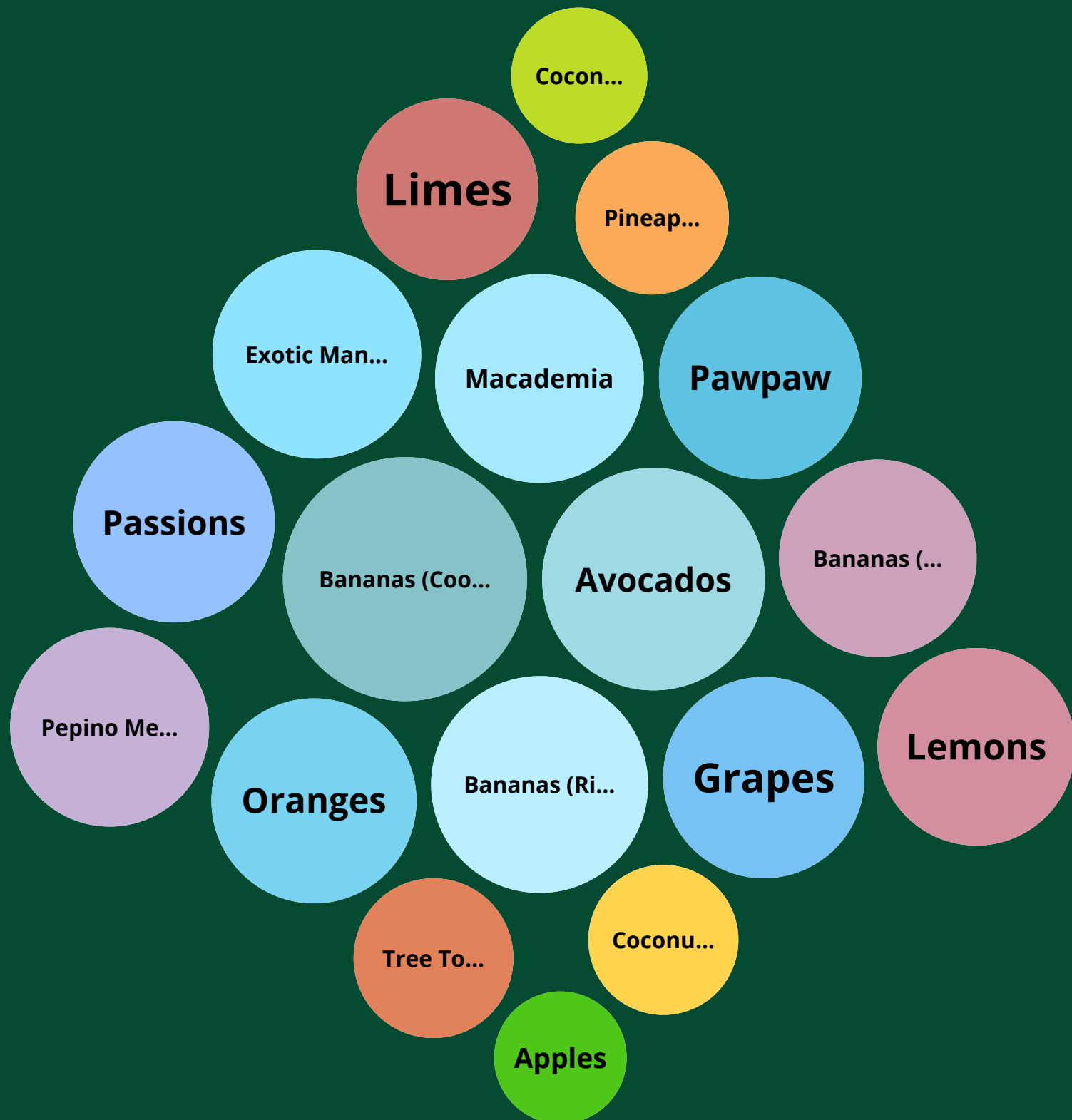
TOP COMMODITIES

HIGH DEMAND COMMODITIES WITH
HIGH PROFIT MARGINS (RANKED)



TOP COMMODITIES

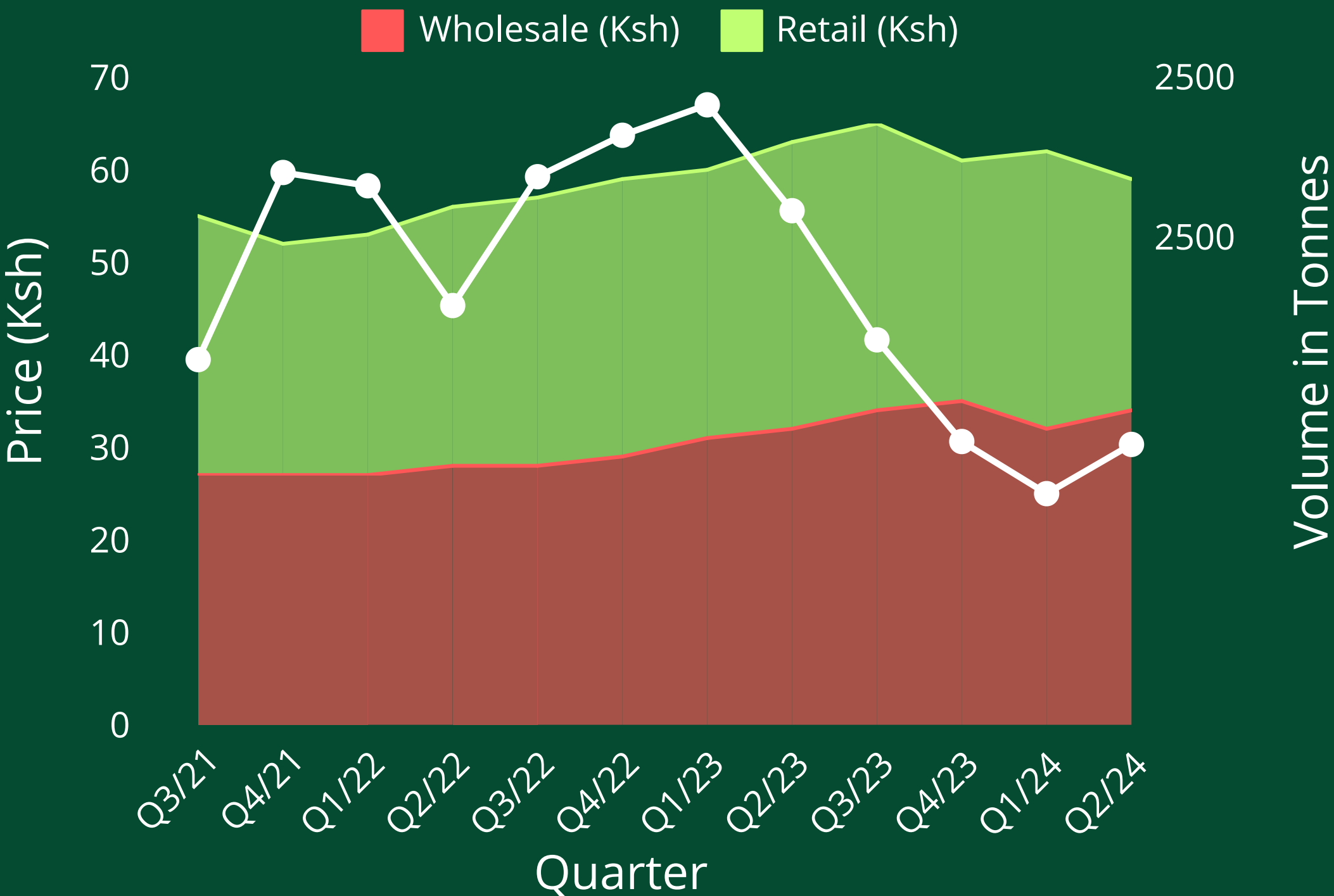
TOP 18 HIGH DEMAND COMMODITIES
WITH HIGH PROFIT % MARGINS



HIGH-DEMAND | HIGH-PROFIT

BANANAS (COOKING)

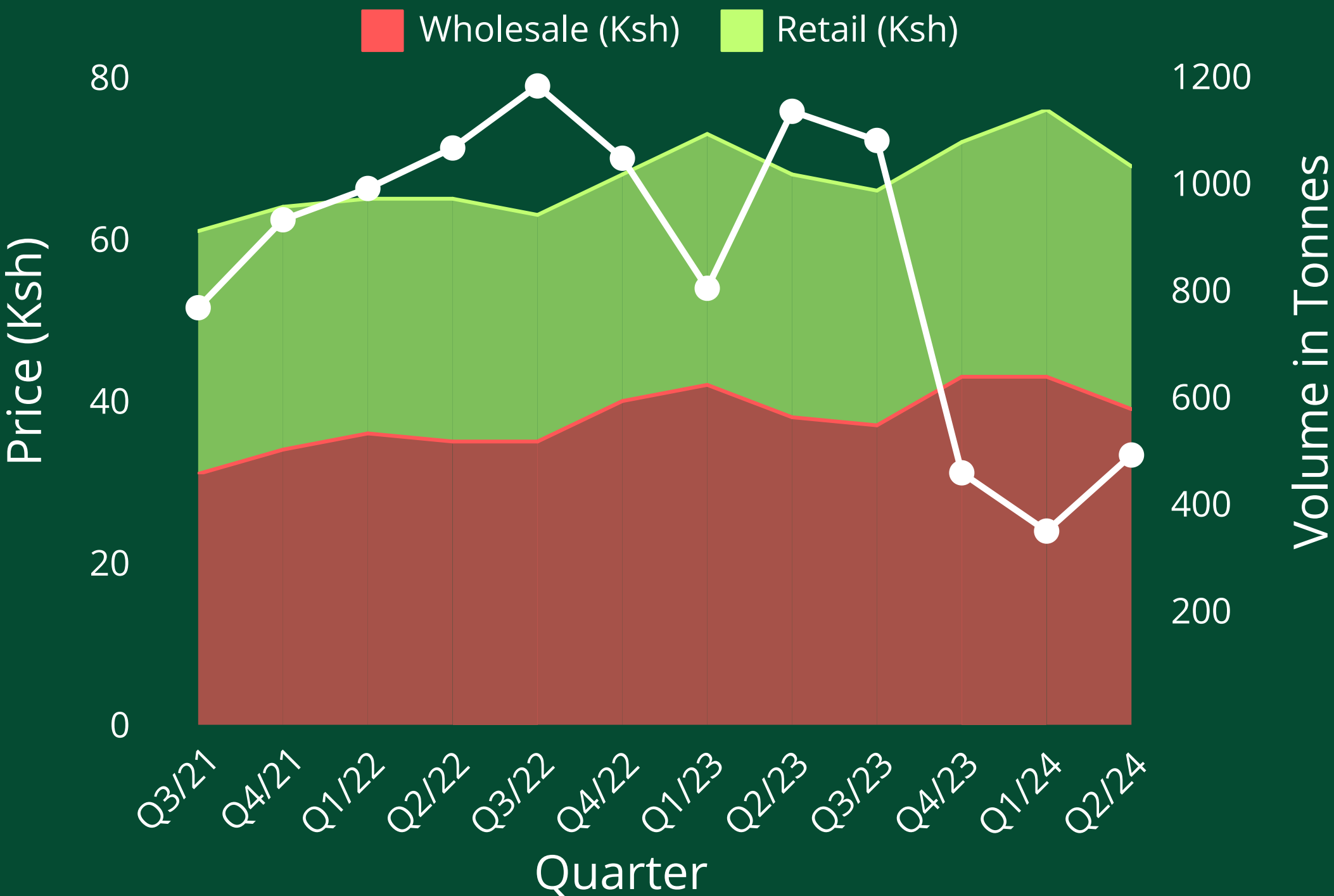
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

AVOCADOS

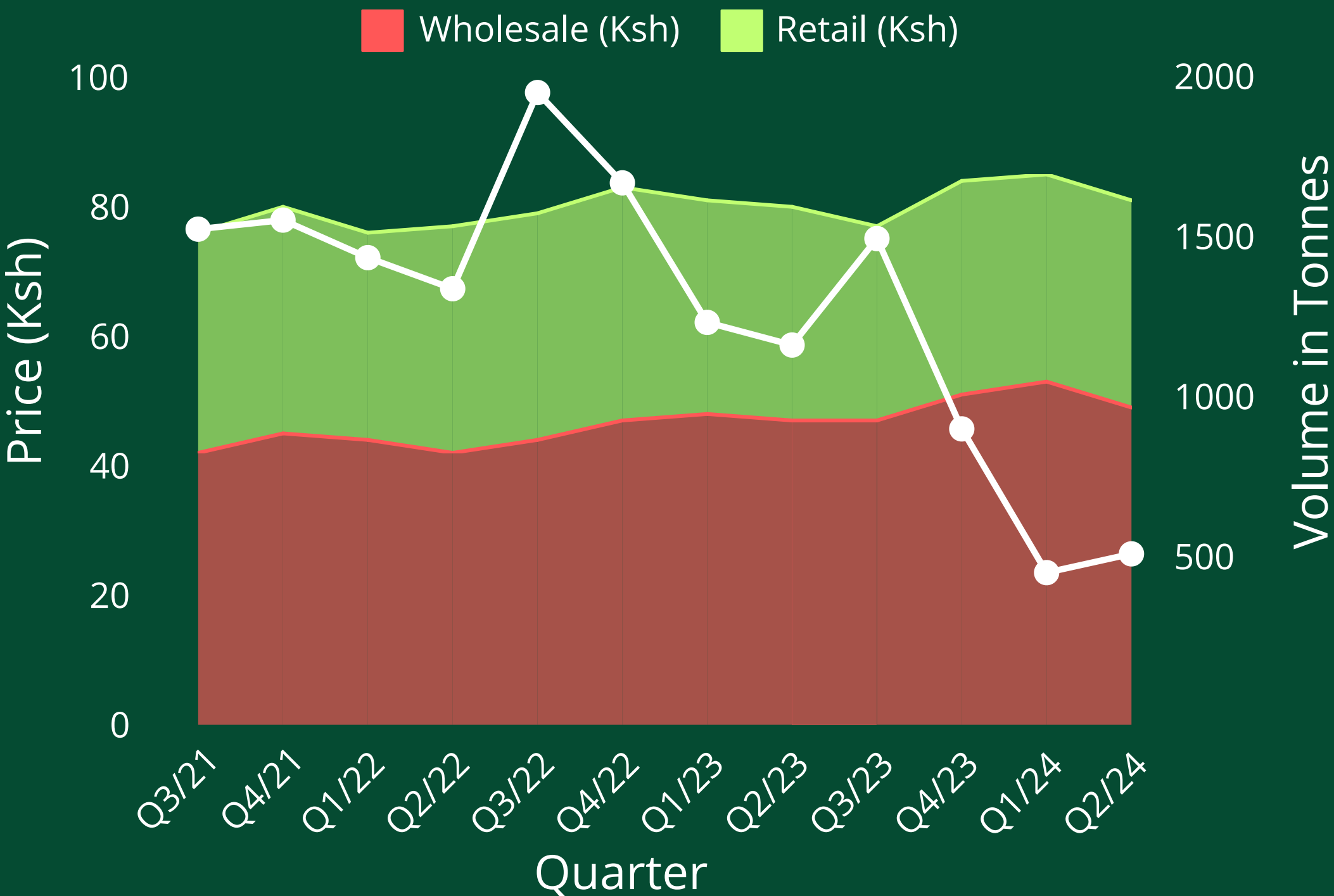
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

ORANGES

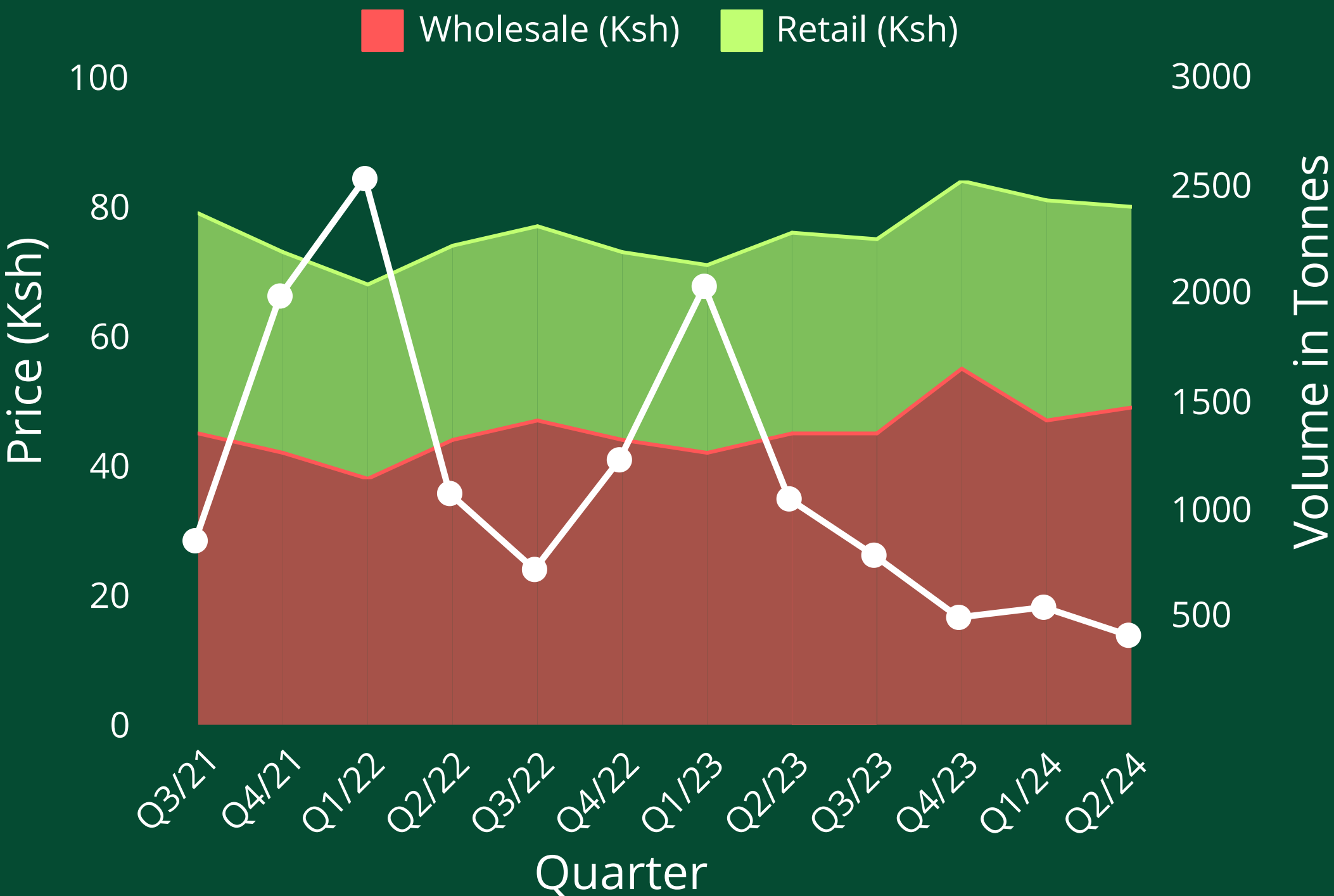
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

EXOTIC MANGOES

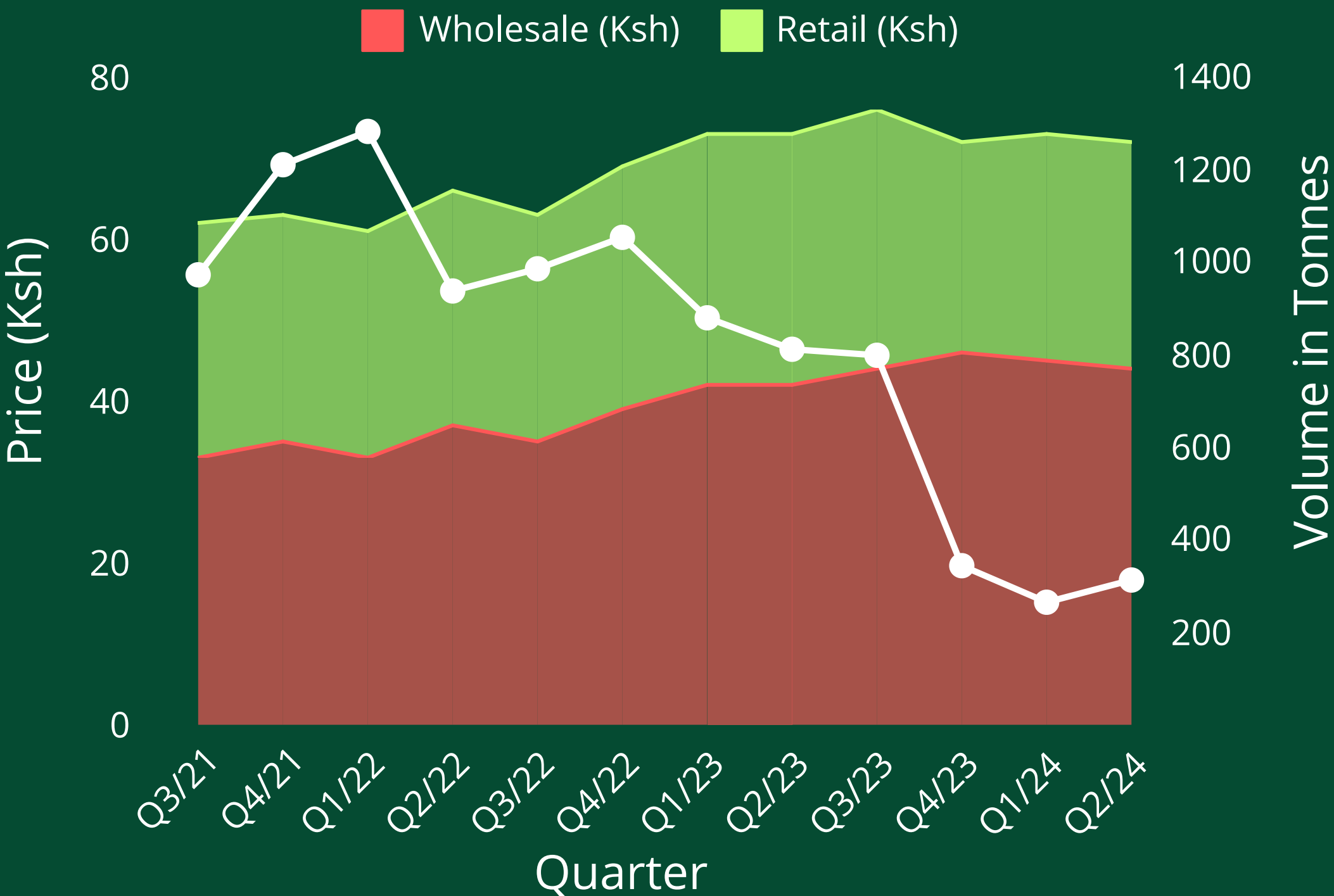
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

BANANAS (RIPE)

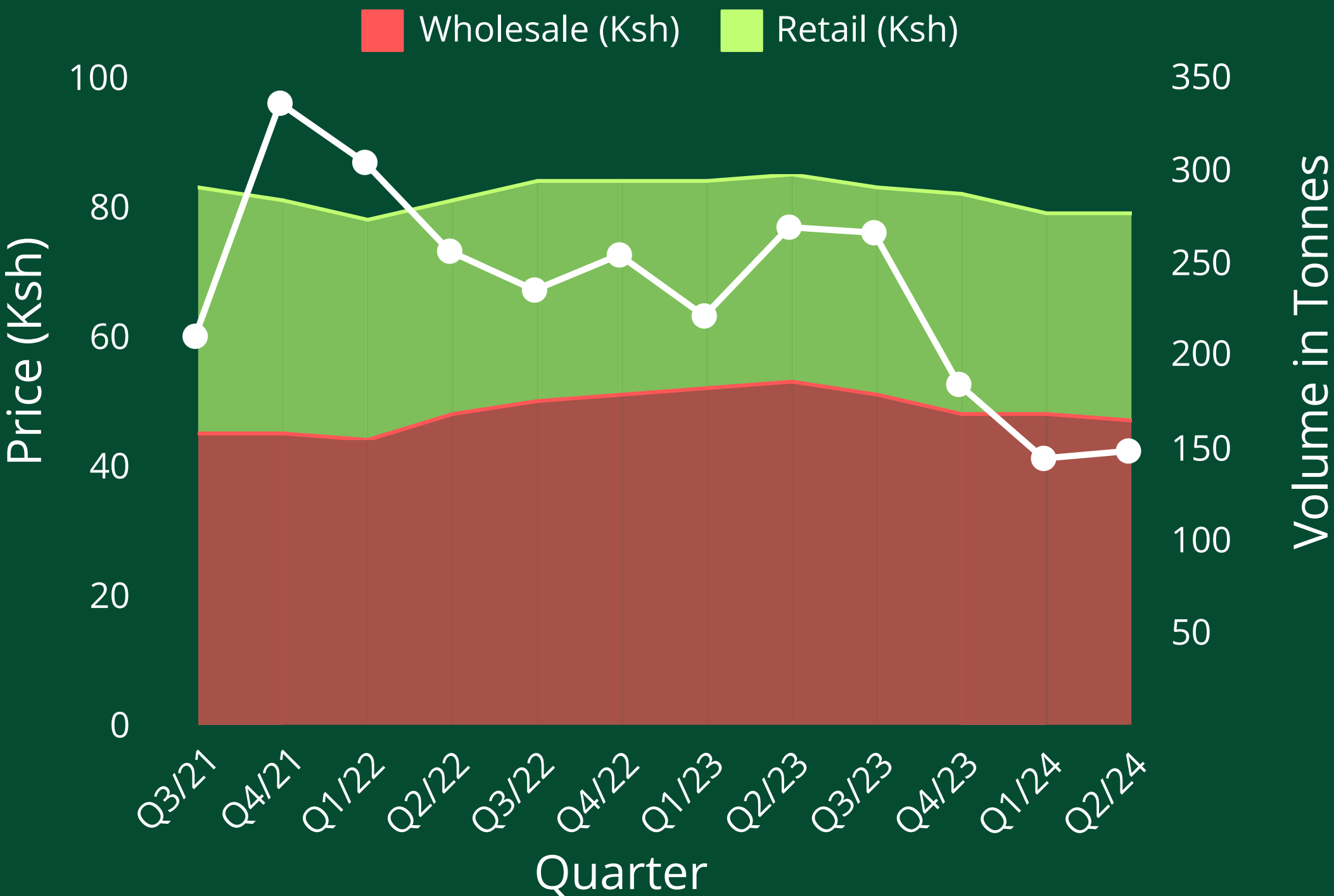
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

PAWPAWS

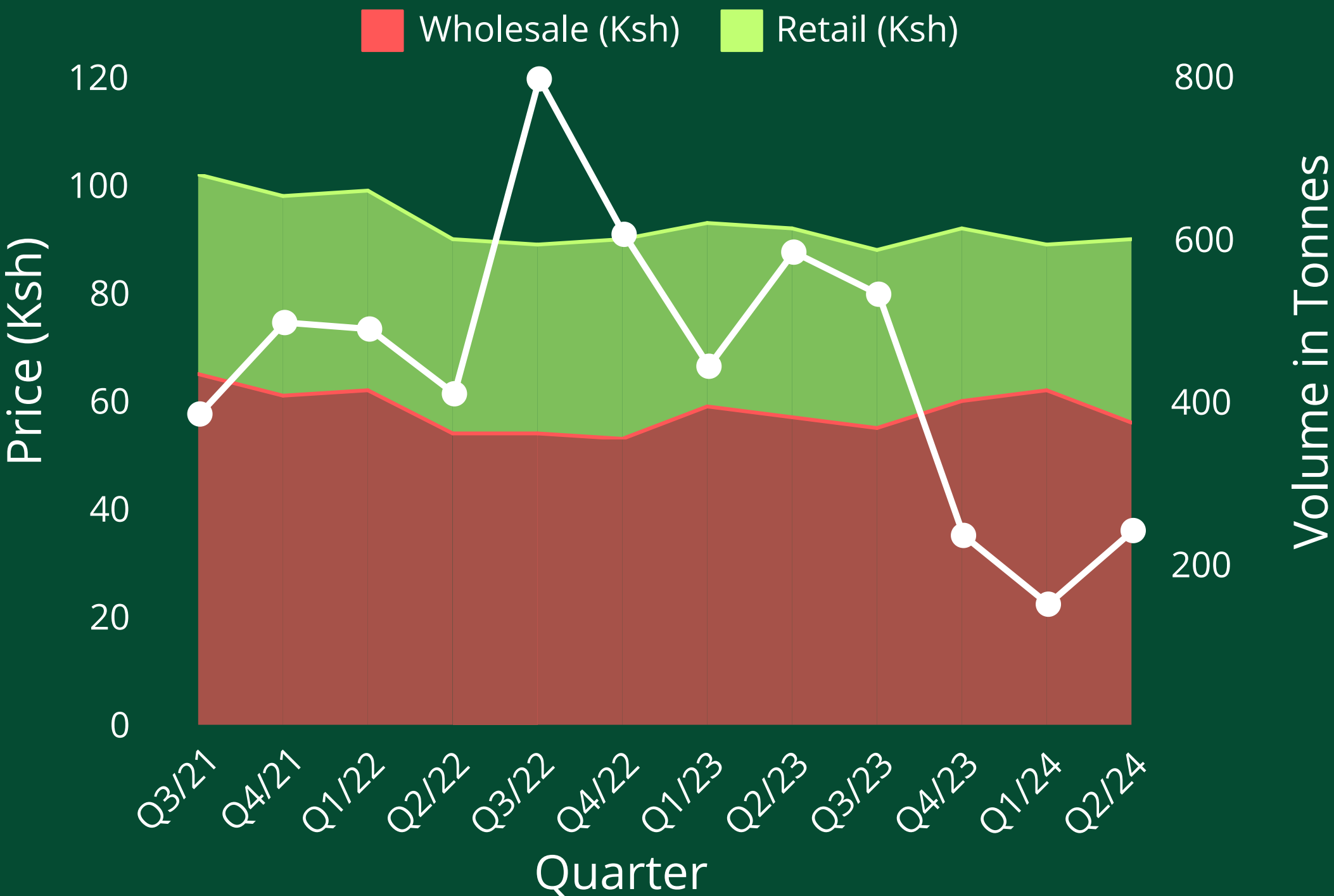
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

LEMONS

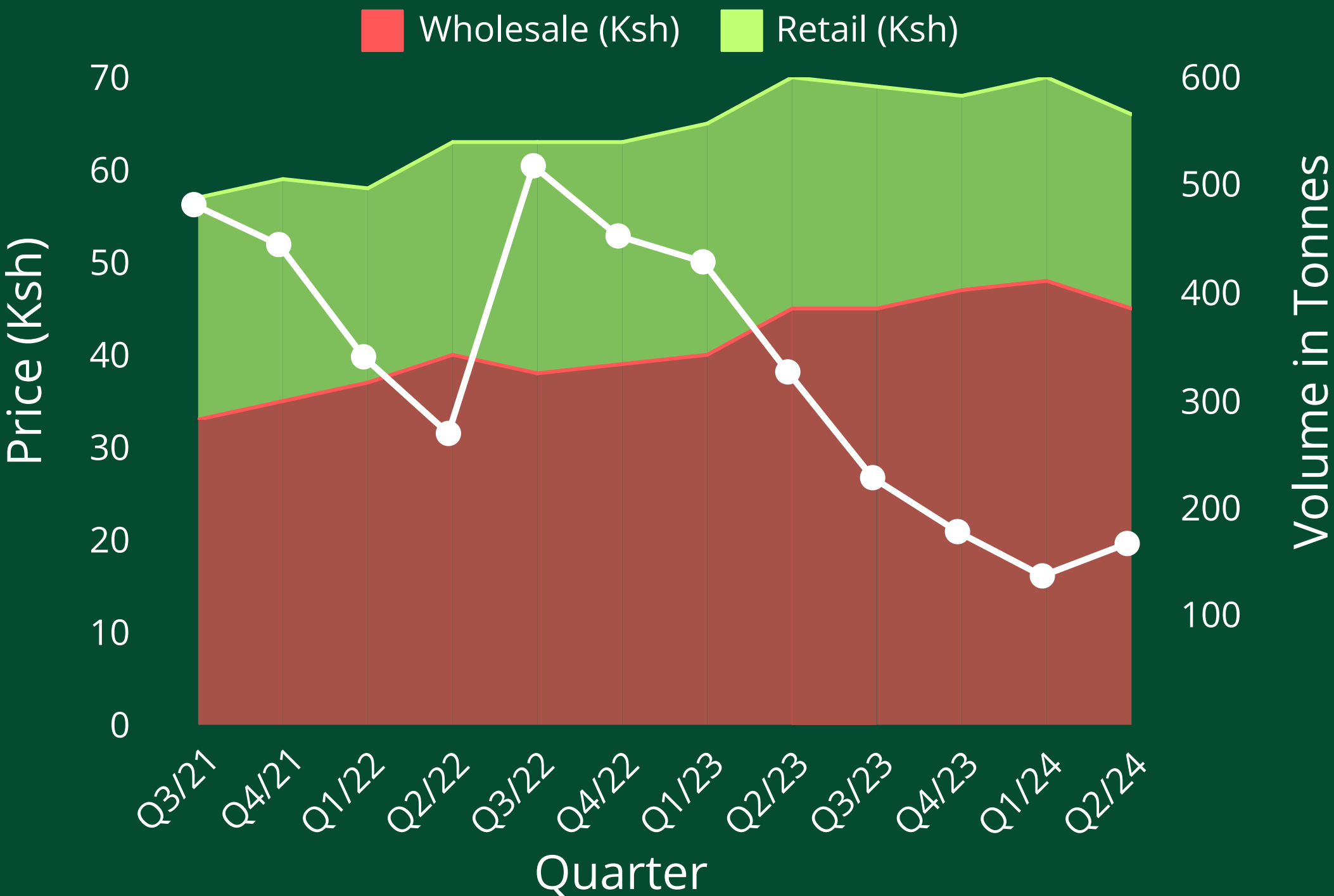
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

BANANAS (PLANTAIN)

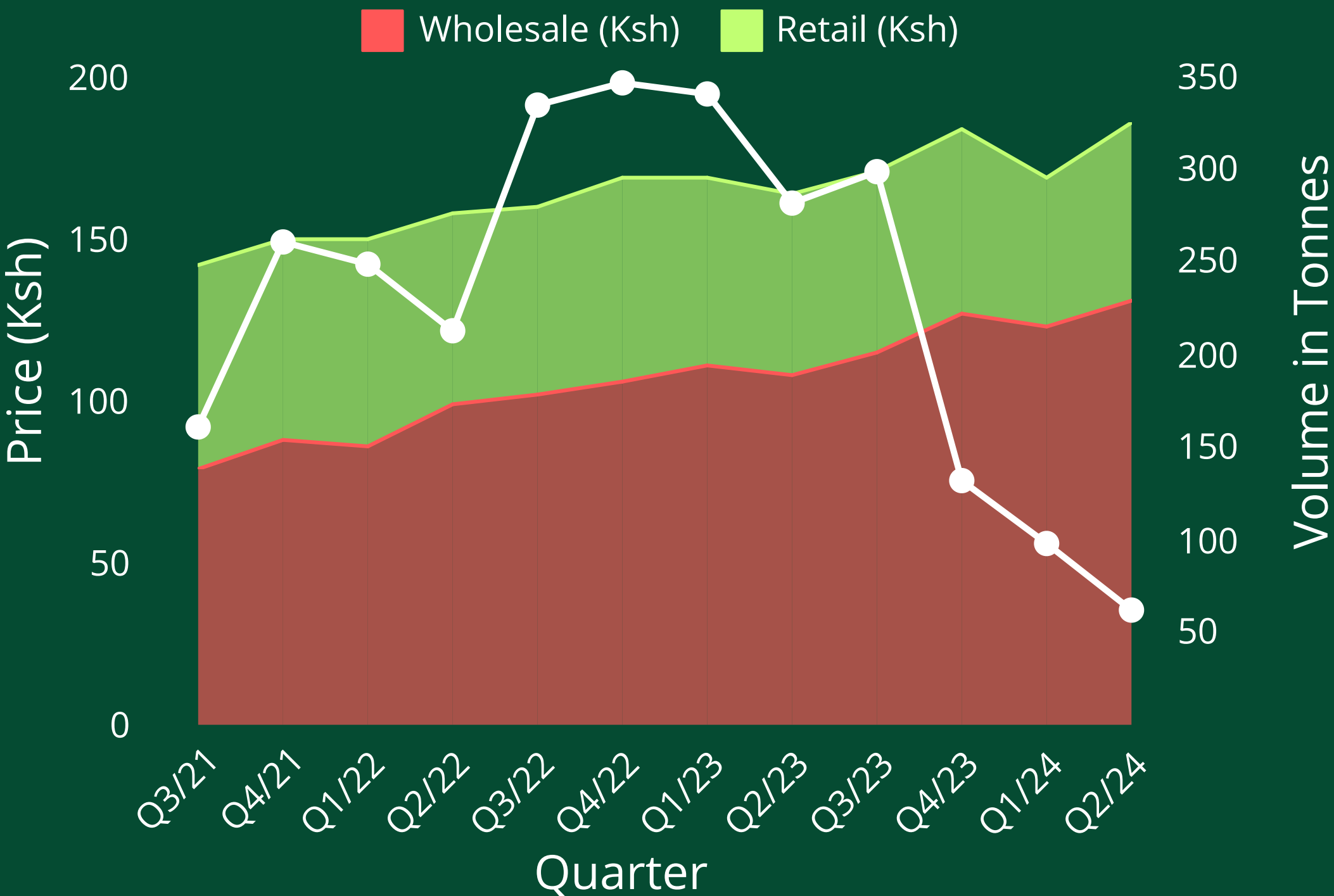
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

PASSIONS

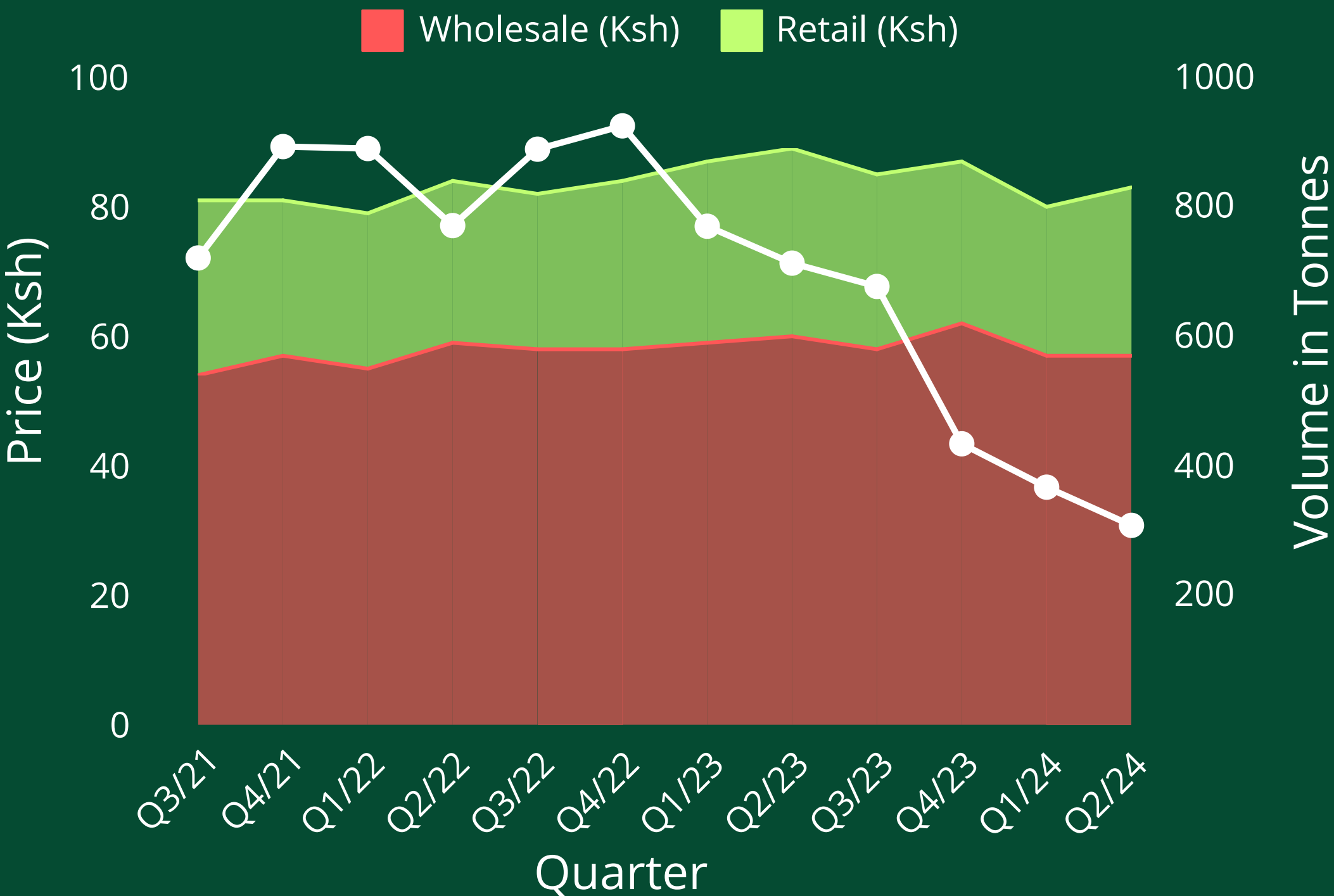
Commodity Pricing Time-Series Analysis



HIGH-DEMAND | HIGH-PROFIT

PINEAPPLES

Commodity Pricing Time-Series Analysis



FINDINGS



There is a high consumer preference for bananas (31%) and citrus fruits (23%).



The top 5 commodities produce 68% of what is in the market, while the top 10 produce 92% of what is in the market.



Predominantly rural (net-production) counties have 91.3% of total volumes sold.



Bananas and Avocados perform very well in terms of demand and resell profit.

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INSIGHTS



Sales are higher in net-production zones, indicating main customers in food markets are brokers and retailers.



92.6% of commodities are retailed between Ksh. 58 - 99, indicating admissible price ranges for mass market options.



2023 was a difficult year across all commodities, with steep declines in volumes sold. However, the trend in 2024 seems to be on the rise.

RECOMMENDATIONS



Smallholder farmers
should consider the
following
high-demand
high-margin
commodities.



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RECOMMENDATIONS



The market is ripe for value-addition cooperatives and farmer support groups, this will stabilize the market all-year round.



Oversupply means under-equipped farmers have tremendous volumes of post-harvest loss. Compost initiatives will greatly benefit smallholder farmers.



Market-linkage solutions will improve smallholder farmer revenues and potentially reduce retail prices for consumers.

CONCLUSION



Hello again! That was quite a lot to read, wasn't it? I'm glad you made it this far.

This is a summary on Part II of my research; ***Perennial Crops***. My next summary will be a deep-dive on Part III: ***Livestock, Fish & Animal Products***.

To catch-up on previous summaries:
Part I: ***Seasonal Crops Summary***
Stay tuned...

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